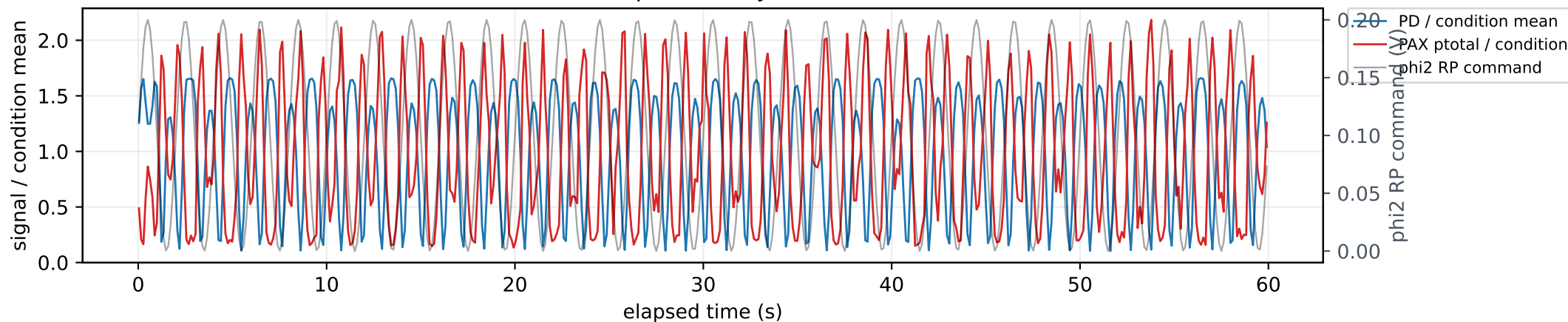
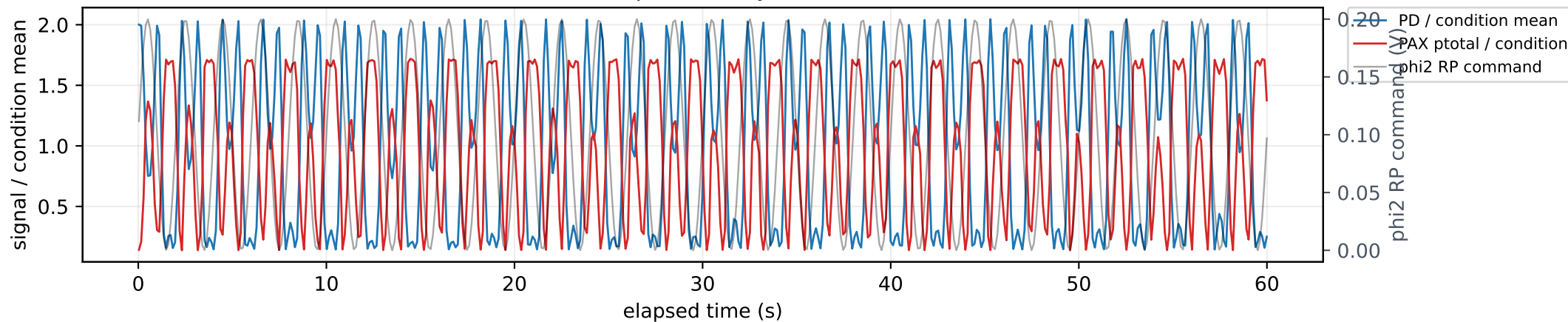


phi2 power balance — time-domain normalized scope traces — 215440_power-balance_upstream-inputs-lon

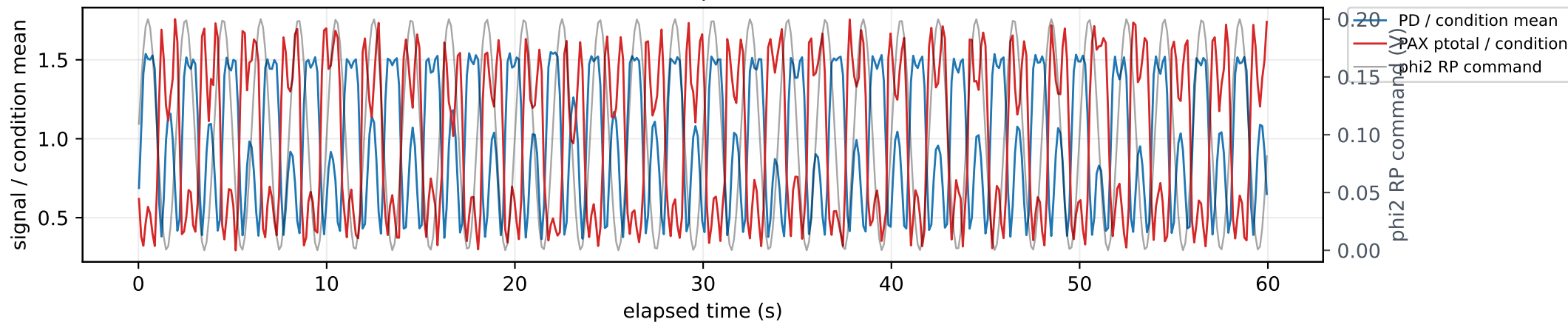
path A only



path B only



both paths



Phi2 power-balance calculations — 215440_power-balance_upstream-inputs-long

condition	samples	mean PD (V)	PD σ (V)	PD contrast (5-95%)	mean PAX ptotal	PAX σ	PAX contrast (5-95%)
path A only	498	0.115674	0.0603	0.8483	6.39659e-05	4.07e-05	0.8401
path B only	499	0.0970895	0.0652	0.8561	0.00016733	9.31e-05	0.8451
both paths	499	0.229494	0.0924	0.5876	0.00023435	0.000114	0.6533

Contrast uses robust 5th/95th-percentile extrema. PD volts and PAX ptotal are different detector units; a physical 40/60 port split requires a relative detector-power calibration.